

MORELIA INTL, Mexico

WMO: 766655

Lat: 19.850N

Lon: 101.026W

Elev: 6033

StdP: 11.76

Time Zone: -6.00 (MEX)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	37.0	39.1	22.7	21.4	62.7	26.5	25.6	63.3	22.0	66.9	19.1	67.6	0.1	180	0.372

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	30.0	89.3	60.3	86.8	59.4	84.6	59.1	65.7	77.4	64.9	76.5	64.5	76.1	6.8	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.8	107.3	67.4	62.2	105.1	67.1	61.1	100.8	66.3	34.2	78.3	33.4	75.6	33.0	76.2	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	17.1	13.6	31.0	93.3	3.4	2.4	28.6	95.0	26.6	96.4	24.7	97.8	22.2	99.5
			28.7	68.2	3.4	2.4	26.2	69.9	24.2	71.3	22.3	72.6	19.8	74.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.8	57.7	60.9	63.8	67.9	71.3	70.5	67.8	67.4	67.0	64.8	60.2	57.9
	DBStd	5.29	2.84	2.96	3.36	2.87	3.00	3.05	2.05	1.92	1.97	3.11	3.37	2.86
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	829	226	117	58	7	1	1	3	3	7	40	145	221
	CDD50	5396	239	306	428	536	659	616	552	540	509	460	307	244
	CDD65	748	0	3	20	93	195	167	90	78	65	35	2	0
	CDH74	9183	133	470	1004	1822	2235	1342	501	460	393	438	251	134
	CDH80	2225	1	48	203	579	849	425	40	28	15	27	9	1
(14) Wind	WSAvg	3.8	3.6	4.1	4.6	4.7	4.6	4.3	3.2	3.1	3.2	3.6	3.1	3.3
(15) Precipitation	PrecAvg	30.4	0.7	0.3	0.3	0.6	1.7	6.0	6.5	6.1	5.4	2.2	0.5	0.2
	PrecMax	38.1	4.4	1.2	1.6	1.6	3.0	12.4	9.5	8.5	9.4	3.6	1.5	1.0
	PrecMin	18.7	0.0	0.0	0.0	0.0	0.0	1.9	3.5	2.6	2.0	0.6	0.0	0.0
	PrecStd	5.2	1.0	0.3	0.4	0.5	0.9	2.2	1.7	1.7	2.0	1.1	0.4	0.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.1	84.3	86.3	89.7	92.8	91.8	82.8	82.6	82.0	82.5	81.0	79.1
		MCWB	55.7	56.7	57.4	59.1	61.7	61.8	63.4	63.9	62.9	61.8	59.2	57.3
	2%	DB	77.2	80.9	84.3	87.7	89.6	88.1	80.8	80.5	80.2	80.6	78.9	77.1
		MCWB	55.4	56.2	57.1	58.6	60.6	60.9	63.6	63.8	63.1	61.6	58.4	56.7
	5%	DB	75.3	79.1	82.4	85.9	87.6	85.6	79.0	78.7	78.5	78.7	77.0	75.4
		MCWB	54.9	55.9	56.7	57.9	60.2	61.3	63.5	63.4	63.0	61.2	58.0	56.2
	10%	DB	73.3	77.1	80.3	84.0	85.6	82.2	77.0	76.9	76.7	76.8	75.1	73.4
		MCWB	54.4	55.5	56.4	57.3	59.7	61.8	63.3	63.6	63.3	61.2	58.1	55.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.1	60.2	61.1	62.8	65.2	66.3	66.6	66.6	66.6	65.6	63.5	61.6
		MCDB	75.1	78.7	80.0	83.1	82.3	80.1	77.9	77.9	77.0	77.0	76.1	75.3
	2%	WB	57.5	58.8	59.7	61.0	63.9	65.0	65.2	65.3	65.2	64.5	61.7	59.2
		MCDB	73.2	77.1	78.7	82.0	80.7	78.1	76.4	76.2	75.6	75.6	74.0	72.6
	5%	WB	56.3	57.4	58.7	59.7	62.9	64.4	64.5	64.6	64.5	63.5	60.3	57.6
		MCDB	72.0	75.3	77.9	81.0	79.8	77.1	75.8	75.4	75.0	74.0	72.5	71.3
	10%	WB	55.3	56.2	57.4	58.7	62.0	63.4	63.7	63.9	63.7	62.6	59.2	56.5
		MCDB	70.7	73.9	76.5	80.0	79.0	75.4	74.4	74.1	73.5	72.7	71.2	70.2
(35) Mean Daily Temperature Range	5% DB	MDBR	32.1	32.6	33.8	33.8	30.0	23.9	21.2	21.2	21.6	26.0	31.6	32.8
		MCDBR	35.6	36.9	37.3	35.5	32.7	29.3	23.7	23.9	24.5	28.8	33.6	35.6
		MCWBR	17.3	16.4	15.8	13.7	11.5	9.7	9.4	9.9	10.3	12.7	16.1	17.7
	5% WB	MCDBR	31.8	32.5	33.0	32.1	28.4	23.6	21.3	21.4	21.1	24.1	29.5	32.2
		MCWBR	16.0	15.1	14.7	13.2	10.1	8.4	9.1	9.4	9.3	11.0	15.0	17.0
(40) Clear-Sky Solar Irradiance	taub		0.329	0.339	0.336	0.375	0.452	0.413	0.400	0.394	0.397	0.372	0.331	0.326
	taud		2.434	2.400	2.406	2.312	2.123	2.270	2.317	2.331	2.328	2.395	2.464	2.468
	Ebn at Noon		301	304	308	294	269	278	281	285	285	289	298	299
	Edh at Noon		34	37	38	42	51	43	42	41	41	36	32	32
(44) All-Sky Solar Radiation	RadAvg		1540	1829	2119	2268	2106	1856	1810	1810	1636	1652	1591	1503
	RadStd		112	165	117	82	123	120	116	96	103	108	119	84

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	N/A	+1.44	-3.11	N/A	-1.33	-1.69	-54	N/A	N/A	N/A

Nomenclature: See separate page